CIF Digitisation Project: Consolidated Report

This report consolidates the field visit findings, the OCR and LLM evaluation note, and the landscaping. It presents the problem context, solution landscape, site observations, implementation progress, challenges, and the next 3 months roadmap for the CIF digitisation effort.

1. Problem Statement

Case Investigation File (CIF) documents are still being handled through a mix of handwritten registers, manual transcription, and portal entry. In practice, this creates avoidable duplication and slows down clinical and field workflows.

The core problem is not only text extraction. It is the end-to-end conversion of noisy, often handwritten, medically sensitive documents into structured records that can be reviewed, validated, and reused safely.

Problem Area	Current Reality	Impact
Record capture	Prescriptions and investigation notes are often handwritten	Hard to read, easy to misinterpret
Duplicate entry	The same data is entered in registers and digital portals	Wasted staff time and repeated errors
Clinical sensitivity	Fields such as diagnosis, dosage, and age are clinically important	Errors can affect care quality and reporting
Operational load	PHC staff already manage patient flow, tests, and reporting	Little room for manual cleanup or re-entry

The project therefore aims to digitise CIF records into a reliable, reviewable, and schema-validated workflow that reduces manual effort and improves data quality.

2. Landscaping

2.1 Industry Overview

Healthcare document digitisation sits at the intersection of OCR, medical record management, multilingual text handling, and human quality control. The source files show a realistic public-health setting:

Industry Characteristic	Why It Matters
Handwritten clinical documents	OCR must handle poor handwriting and inconsistent formatting
Multilingual notes	The pipeline must support Hindi and mixed-language content
Safety-critical fields	Errors in dosage, diagnosis, age, or date require human review
Public health reporting	Outputs need to be structured enough for downstream dashboards and registers
Privacy and compliance	The system must be designed with access control and data minimisation in mind

This is why a single-model approach is usually not enough. The better industry pattern is a hybrid pipeline with OCR, LLM structuring, validation, and review.

2.2 Multi-Model Approach

The landscaping study recommends a layered architecture rather than a single model doing every task.

Approach	Strengths	Limitations	Fit for CIF
OCR only	Fast, cheap, and simple	Weak on handwriting, partial scans, and clinical structuring	Useful as the first extraction layer, not sufficient alone
LLM only	Can infer structure and clean text	More expensive, more variable, and less ideal for raw image interpretation	Not preferred as the only solution
Hybrid OCR + LLM	Separates extraction from structuring	Requires orchestration and validation	Best overall fit for CIF digitisation
Multi-model hybrid	Allows fallback, translation, and quality tiers	More complex to operate	Best for production and scaling

Recommended model roles:

Pipeline Layer	Candidate Models / Tools	Primary Role
OCR / document intelligence	Sarvam Vision	Extract text from scanned CIF documents
Post-OCR structuring	Gemini 2.5 Flash-Lite, GPT-4o mini, GPT-4.1 mini	Convert masked OCR text into structured records
Higher-end reasoning	GPT-4.1, Claude Sonnet 4	Hard-case reasoning and quality benchmarking
Sarvam-native alternatives	Sarvam 30B, Sarvam 105B	Keep more of the workflow inside the Sarvam stack
Multilingual normalisation	Sarvam Translate, Mayura	Handle Hindi or mixed-language output
Offline baseline	Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-32B	Local or air-gapped fallback after masking
Routing / abstraction	OpenRouter	Multi-provider access and fallback orchestration

The practical takeaway is that the extraction system should be modular. OCR should be swappable, the structuring LLM should be benchmarked independently, and the final output must pass a strict validation layer before it is accepted.

2.3 Evaluation Methodology

The evaluation note and landscaping study both point to the same principle: compare candidates on the same test set, under the same conditions, and score them on both quality and operational cost.

Metric	How It Is Measured	Why It Matters
Field accuracy	Exact or normalised match against human-reviewed ground truth	Core measure of correctness
Completeness	Percentage of required fields populated correctly	Important for usable structured records
Robustness	Repeated runs with small prompt or version changes	Important because LLMs are probabilistic
Concordance	Model-to-model and model-to-human agreement	Identifies stable candidates
Latency	End-to-end time and model inference time	Important for PHC throughput
Token usage	Input, output, and total tokens per document	Drives LLM cost and prompt design
Cost per document	OCR cost plus LLM and infrastructure cost	Needed for deployment planning
Failure modes	Handwriting, blur, mixed script, partial scans, missing values	Helps design fallback and review rules

Recommended evaluation design:

Step	Evaluation Activity	Output
1	Build a fixed benchmark set of representative documents	Stable comparison dataset
2	Run each candidate model or model stack on the same set	Comparable outputs
3	Score field-level correctness for key clinical fields	Accuracy report
4	Measure latency and token usage	Cost and throughput report
5	Review edge cases manually	Error taxonomy and mitigation ideas
6	Shortlist the best stack by quality-cost tradeoff	Production recommendation

2.4 Pricing Analysis

The pricing note assumes 7,000 to 7,500 documents per year, one page per document for budgeting, and approximately 2,000 input tokens plus 250 output tokens per document after OCR and masking.

OCR / Document Intelligence Pricing

Service	Pricing Signal	Cost per Document	Annual Cost for 7,000 Docs	Annual Cost for 7,500 Docs
Sarvam Document Intelligence / Vision	Rs 0/page in API docs, Rs 1.50/page in the Sarvam pricing page	Rs 0 to Rs 1.50	Rs 0 to Rs 10,500	Rs 0 to Rs 11,250

LLM Structuring Pricing

Model	Pricing Basis	Approx. Cost per Document	Annual Cost for 7,000 Docs	Annual Cost for 7,500 Docs
Gemini 2.5 Flash-Lite	USD 0.10 input / USD 0.40 output per 1M tokens	USD 0.00030	USD 2.10	USD 2.25
GPT-4o mini	USD 0.15 input / USD 0.60 output per 1M tokens	USD 0.00045	USD 3.15	USD 3.38
Gemini 2.5 Flash	USD 0.30 input / USD 2.50 output per 1M tokens	USD 0.00123	USD 8.58	USD 9.19
GPT-4.1 mini	USD 0.40 input / USD 1.60 output per 1M tokens	USD 0.00120	USD 8.40	USD 9.00
Claude Haiku 3.5	USD 0.80 input / USD 4.00 output per 1M tokens	USD 0.00260	USD 18.20	USD 19.50
GPT-4.1	USD 2.00 input / USD 8.00 output per 1M tokens	USD 0.00600	USD 42.00	USD 45.00
Claude Sonnet 4	USD 3.00 input / USD 15.00 output per 1M tokens	USD 0.00975	USD 68.25	USD 73.13

Planning Summary

Pipeline Option	OCR Cost	LLM Cost	Deployment Style	Best Use Case
Sarvam Vision + Gemini 2.5 Flash-Lite	Rs 0 to Rs 11,250 / year	USD 2.10 to USD 2.25 / year	API-only	Lowest-cost API pipeline
Sarvam Vision + GPT-4o mini	Rs 0 to Rs 11,250 / year	USD 3.15 to USD 3.38 / year	API-only	Very cost-effective and simple
Sarvam Vision + GPT-4.1 mini	Rs 0 to Rs 11,250 / year	USD 8.40 to USD 9.00 / year	API-only	Balanced cost and quality
Sarvam Vision + Claude Haiku 3.5	Rs 0 to Rs 11,250 / year	USD 18.20 to USD 19.50 / year	API-only	Stronger quality benchmark
Sarvam Vision + Sarvam 30B	Rs 0 to Rs 11,250 / year	Pricing page says free per token	API or local runtime	Sarvam-native balance
Sarvam Vision + Sarvam 105B	Rs 0 to Rs 11,250 / year	Pricing page says free per token	API or server-centric H100 runtime	Highest quality Sarvam-native option
Sarvam Vision + Qwen2.5-7B local	Rs 0 to Rs 11,250 / year	USD 17.50 to USD 18.75 / year equivalent	Local GPU	Offline or controlled deployment

The field visit also matters for pricing. Since PHCs already have only a small number of desktops and the current process is manual, an API-first stack is more practical than a heavy on-prem GPU deployment. The main savings come from removing duplicate manual entry and reducing rework.

3. Situational Analysis

3.1 Field Visit Details

Item	Details
Site	Urban Primary Health Centre in Lingarajapuram, Bengaluru
Study region	Lingarajapuram, Bengaluru, Karnataka
Visit date	17-04-2026
Purpose	Understand the real operational requirements for CIF digitisation

The field visit confirmed that the PHC workflow is highly structured operationally, but record capture is still largely manual. The people, registers, tests, and portal forms already exist. What is missing is a clean digital capture layer at the point of care.

3.2 Staff Roles and Information Flow

Role	Responsibility	Main Data Handled
Medical Officer	Clinical authority and final decision-making	Diagnosis, prescription direction, treatment oversight
Staff Nurse	Primary patient documentation and Treatment	Name, age, gender, date, disease, treatment
Pharmacist	Medicine dispensing	Issued medicines and injections
Lab Technician	Diagnostic testing	Test data and lab registers
PHC Officer	Supervisory consolidation	Primary, diagnostic, and regional reporting access
Health Investigating Officer	Ward-level surveillance and public health action	Local outbreaks, drainage issues, quarantine-related observations
D Group	Crowd control and cleaning support	Facility cleanliness and patient flow support
ASHA workers	Community-level reporting	Pregnant women, disabled persons, and household-level data

3.3 Registers, Portal, and Forms

Record System	Who Uses It	What It Stores	Current Limitation
Handwritten register - staff nurse	Staff nurse	Primary patient details and prescriptions	Time-consuming and error-prone
Handwritten register - lab technician	Lab technician	Diagnostic data	Separate from primary records
IHIP portal	Staff nurse, lab technician, PHC officer	Daily digital reporting	Requires repeat manual entry
P Form	Staff nurse	Primary data entry	Same information is also written elsewhere
L Form	Lab technician	Diagnostic entry	Not directly linked to prescription capture
S Form	PHC officer	Supervisory and regional data	Higher-level access but still dependent on manual input

3.4 Work Culture and Real-World Workflow

Clinical workflow

Step	Workflow Stage	Description
1	Patient arrival	Patient arrives at PHC
2	Examination	Examination by Medical Officer / Staff Nurse
3	Record capture	Patient details recorded in handwritten register
4	Dispensing	Medicines issued by Pharmacist
5	Diagnostics	Lab tests conducted if required
6	Digital entry	Data entered into IHIP portal

Community and reporting workflow

Step	Workflow Stage	Description
1	Household data collection	ASHA workers collect household data
2	Ward monitoring	Health Investigating Officers monitor ward-level events
3	Consolidation	PHC Officer consolidates reports
4	Record alignment	Data aligned with PHC records

This workflow shows that the PHC process is already operationally organized, but the capture layer is fragmented across paper and digital systems. Patient information moves from consultation to dispensing, then to diagnostics, and finally into reporting. Because the same data is written more than once, transcription burden and inconsistency risk remain high. A digitisation layer can reduce duplication while preserving the existing clinical workflow.

3.5 Mother Card, RCH ID, and Continuity of Care

Feature	Purpose	Why It Matters
Mother Card	Given to pregnant women during antenatal care	Supports follow-up across pregnancy
RCH ID	Unique identifier for a pregnant woman	Helps track mother and child health over time
Child vaccination linkage	Tracks the child up to the 16th year vaccination cycle	Requires accurate and persistent records

3.6 Tests and Referral Structure

Test Category	Examples Observed	Notes
Vector-borne disease testing	Dengue NS1, malaria rapid testing, malaria smear	Common in the local disease burden
Maternal health	ANC checkups, haemoglobin, serology, urine tests	Important for mother card and RCH continuity
Diabetes screening	OGCT, GRBS, FBS, PPBS	Routine non-communicable disease monitoring
Fever and infection workup	MR swab, Typhoid Vidal test	Used for symptomatic diagnosis
Referral-based testing	Thyroid and other unavailable tests	Referred to hospitals when local tools are missing

3.7 Infrastructure and Constraints

Constraint	What Was Observed	Why It Matters for Design
Limited desktops	PHCs generally have only a few desktops	The solution should be lightweight and browser-friendly
No prescription capture tool	Details are not captured at the source digitally	OCR digitisation can remove the first manual copy step
Double entry burden	Data is written in registers and then re-entered in IHIP	Automation can save time and reduce errors
Handwritten source data	Registers are still handwritten	OCR must be robust to noise and variation
Local disease mix	Vector-borne diseases such as malaria and dengue are common	The system should support frequent, repetitive case types
Referral dependence	Some tests are not available on-site	The system should support incomplete local test coverage

4. Work Done So Far

4.1 System Architecture

The current implementation already contains a working digitisation platform rather than only a concept note.

Layer	Implemented Component	Function	Status
Access control	Auth0 based sign-in and role mapping	Controls who can access each workflow	Implemented
Frontend	React and MUI application	Upload, processing, case review, dashboard, and reports screens	Implemented
Backend API	FastAPI async job service	Accepts uploads and manages digitisation jobs	Implemented
Model access	OpenRouter integration	Routes multimodal extraction requests to the selected model	Implemented
Extraction model	anthropic/claude-sonnet-4.6	Multimodal extraction for structured CIF output	Implemented in current stack
Storage	Runtime job state and document storage. Used SQLlite to store the primary details of the patient and s3 to store uploaded medical prescription	Supports job tracking and file handling	Implemented
Validation	Pydantic models and review rules	Applies schema and business-rule checks	Implemented
Reporting	Dashboard and results views	Displays extracted records and review status	Implemented

4.2 Implementation Steps Completed

Step	Stage	What the Current System Does
1	Authentication	User signs in through Auth0 and receives a role-based route set
2	Upload	User uploads a CIF document from the Upload page, with preview support
3	Job creation	Backend creates a digitisation job through `/api/digitize`
4	File validation	Backend validates MIME type, content, and request payload
5	Extraction	OpenRouter forwards the document to the selected multimodal model
6	Normalisation	Backend cleans values, translates where needed, and standardises outputs
7	Quality control	Schema checks and validation rules flag uncertain records
8	Review	Case Review page supports manual edits and verification
9	Reporting	Dashboard and reports screens surface the latest records and session history

4.3 Models and Tools

Tool / Model	Role in the Project	Current Use
OpenRouter	Multi-provider routing layer	Used for model access and abstraction
Claude Sonnet 4.6	Multimodal extraction model	Current production-style extraction model in the app
Sarvam Vision	OCR/document intelligence candidate	Used in landscaping and pricing comparison
Gemini 2.5 Flash-Lite	Low-cost LLM candidate	Used in pricing and evaluation comparison
GPT-4o mini	Low-cost LLM candidate	Used in pricing and evaluation comparison
GPT-4.1 mini	Stronger low-cost LLM candidate	Used in pricing and evaluation comparison
Claude Haiku 3.5 / Claude Sonnet 4	Higher-quality benchmark models	Used for comparative landscaping
Sarvam 30B / 105B	Sarvam-native structured reasoning candidates	Used for alternate stack planning
Qwen2.5 family	Local fallback models	Used for offline and cost-sensitive planning
Pydantic	Schema validation	Enforces structured records and QC checks
FastAPI	Backend orchestration	Manages upload and processing endpoints
React + MUI	Frontend interface	Provides upload, review, dashboard, and reports screens
S3	File storage	Stores uploaded prescription files

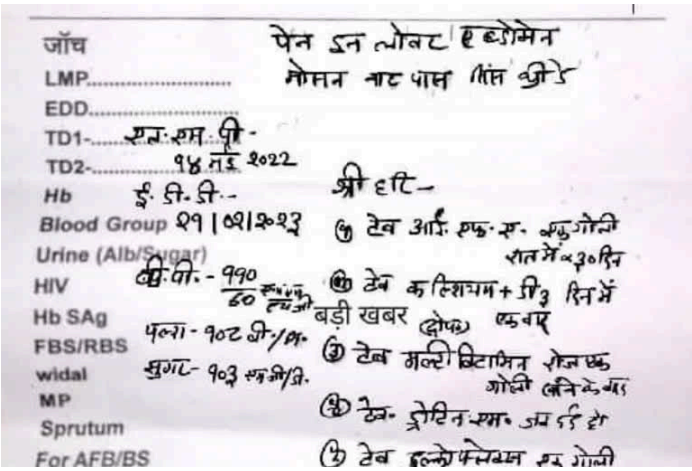
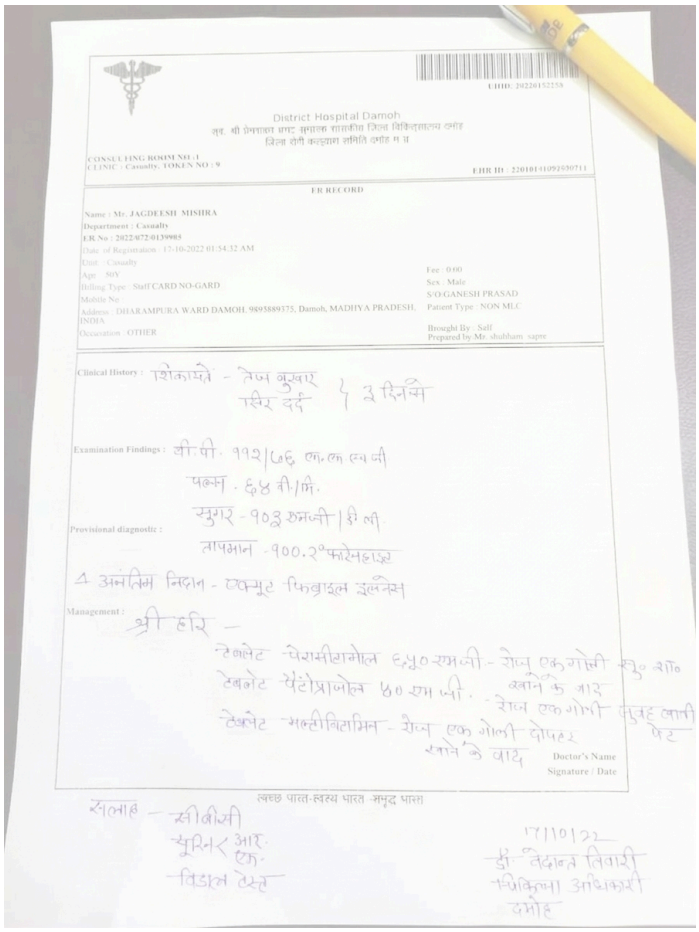
4.4 Data Pipeline

Step	Pipeline Stage	Description
1	Document Upload	Document is uploaded into the system
2	Backend Validation	Backend validates the request and payload
3	Storage	Document is stored for traceability and job tracking
4	Extraction	Multimodal model extracts text and fields
5	Normalisation	Extracted output is cleaned and standardised
6	Validation	Schema and business rules are applied
7	Human Review & Correction	Reviewer checks and corrects uncertain fields
8	Reporting Dashboard	Final structured data is shown in the dashboard

This pipeline ensures that document handling stays traceable from upload to reporting. The backend first validates the file, then stores it, then sends it through extraction and normalisation. After that, schema rules filter out incomplete or uncertain results so the reviewer only sees records that need human attention. The final output is made visible in the dashboard and results views for operational use.

4.5 Illustrations

The following figures show real sample inputs that motivated the design. They are useful because they demonstrate why a robust extraction pipeline must handle brightness differences, handwriting variation, and document quality issues.

Figure	Sample	What It Shows
Figure 1		A handwritten clinical document with mixed typed and handwritten content
Figure 2		A noisy sample that reflects poor scan quality and capture variation
Figure 3		A brighter sample that helps test extraction under cleaner conditions

4.6 Visual Workflow Summary

Layer	Description
Capture	Document is uploaded from the frontend
Extraction	Multimodal model reads the document
Structuring	Output is converted into a schema-aligned record
Validation	Rules flag missing, invalid, or uncertain fields
Review	Human reviewer can correct and confirm
Reporting	Result is surfaced in the UI and preserved for analysis

4.7 OCR Improvement Using Data Augmentation

OCR struggles in this project because the source documents are highly variable. Handwriting differs across doctors, scans may be blurred or skewed, lighting may be uneven, and real-world capture often introduces noise, shadows, and low contrast. These issues reduce OCR confidence and make downstream structuring less reliable.

Data augmentation can make the OCR stage more robust by exposing the model or training pipeline to realistic document distortions before deployment.

Technique	Purpose	Example Impact
Noise injection	Simulates real-world scans	Improves robustness
Blur augmentation	Handles low-quality images	Better extraction from poor scans
Rotation/skew	Handles misaligned documents	Improves detection accuracy
Contrast/brightness changes	Handles lighting variations	Stabilises OCR output
Synthetic handwriting variations	Improves generalisation	Better handling of doctor handwriting

These augmentation methods help the OCR system learn to tolerate the kinds of document imperfections seen during the field visit and in sample prescriptions. They are especially useful when the training or evaluation set is small, because they expand the effective variety of documents without requiring new manual annotation. In this project, augmentation can improve both raw text extraction and the reliability of the structured fields that depend on that OCR output.

5. Challenges

Challenge Type	Specific Challenge	Project Impact
Technical	Handwriting variation across doctors and facilities	OCR quality can vary significantly
Technical	Blur, shadows, folds, and partial scans	Increases extraction errors and missing fields
Technical	Mixed-language and shorthand content	Requires translation and normalisation
Data-related	Limited labeled ground truth	Hard to benchmark models fairly
Data-related	Inconsistent field formats across documents	Makes schema alignment difficult
Data-related	Duplicate or unreadable entries	Creates ambiguity in validation
Deployment	PHCs have only a few desktops	Solution must be light, browser-based, and easy to use
Deployment	Manual workflows are already busy	Human review must be minimal and targeted
Deployment	API and model cost uncertainty	Requires careful benchmarking and budget control
Deployment	Privacy and compliance requirements	Access control, retention, and auditability matter

6. Future Work (Next 3 Months Deliverables)

6.1 Roadmap Overview

Month	Milestone	Key Deliverables	Expected Outcome
Month 1	Benchmark foundation	Final schema, annotation guide, labelled test set, baseline model runs	Stable evaluation dataset and first accuracy baseline
Month 2	Model comparison and hardening	Comparative benchmarks, prompt refinement, confidence routing, QC rules, error taxonomy	Shortlisted model stack and improved reliability
Month 3	Pilot-ready integration	End-to-end integration, reviewer workflow, deployment plan, SOPs, final report	Pilot-ready release and handover package

6.2 Expected Improvements

Improvement Area	Target Direction
Model quality	Higher field-level accuracy on handwritten and mixed-language documents
Manual effort	Less duplicate typing and fewer correction cycles
Cost	Lower cost per document through better model selection and token control
Scalability	Easier expansion across PHCs through a browser-based, API-driven workflow
Reliability	Better handling of low-quality scans and uncertain fields
Governance	Stronger validation, review, and audit support

Closing Note

The combined evidence from the field visit, pricing study, and landscaping work shows that CIF digitisation is best treated as a full operational pipeline, not just an OCR task. The right solution must capture documents, structure the content, validate the result, and support review with minimal burden on PHC staff.